Dori Dhore Maro Taan

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Problem 1. Let $\triangle ABC$ be an acute triangle with $BC > AC$. Let S be the centroid of triangle ABC and let F be the foot of the perpendicular from C to side AB. The median CS intersects the circumcircle γ of triangle $\triangle ABC$ at a second point P. Let M be the point where CS intersects AB. The line SF intersects the circle γ at a point Q, such that F lies between S and Q. Prove that the points M, P, Q and F lie on a circle.

Problem 2. Let α be a non-zero real number. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$xf(x+y) = (x+\alpha y)f(x) + xf(y)$$

for all $x, y \in \mathbb{R}$.

Problem 3. A square has been divided into 2022 rectangles with no two of them having a common interior point. What is the maximal number of distinct lines that can be determined by the sides of these rectangles?

Problem 4. Prove that there exists infinitely many positive integers n for which the equation

$$x^2 + y^{11} - z^{2022!} = n$$

has no solution (x, y, z) over the integers.